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Western Cape Government
FOR YOU

EDUCATIONAL PROGRAM REPORTING SYSTEM (EPRS)

INF3003W User Requirement Specification

9 August 2026

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Western Cape Government

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Abbreviations Table

Abbreviation	Full Term
ASP.NET	Active Server Pages Network Enabled Technologies
Cel	Centre for e-Innovation
C#	C Sharp
CSV	Comma-separated values
DBMS	Database Management System
DHET	Department of Higher Education and Training
DSD	Department of Social Development
EPRS	Educational Programme Reporting System
IT	Information Technology
KPI	Key Performance Indicator
MVC	Model, View, and Controller
POPIA	Protection of Personal Information Act
RAG	Red, Amber, Green
SQL	Structured Query Language
SMD	State Machine Diagram
URS	User Requirement Specification
UML	Unified Modelling Language
WCG	Western Cape Government

Table 1: List of Abbreviations

Introduction

This document presents the User Requirement Specification for the proposed Educational Program Reporting System (EPRS), developed by Team 8 (Wink) for the INF3000W course. The purpose of this document is to define the functional requirements, non-functional and user roles required to transition the Western Cape Government's juvenile facility from error-prone manual data logs to an automated academic management platform. This URS establishes a definitive technical agreement on the system's operational scope. It outlines exactly how staff members will capture academic milestones, safely manage student records, distribute report cards and track system audits.

Project Definition Statement

Group Name: Wink	**Group Leader:** Nicole Scott-Emuakpor (Project Manager)
Sponsor name: Daneel Carolissen (Western Cape Government)	**Contact person:** Daneel Carolissen
Date: 18/06/2026	**Project Title:** Educational Program Reporting System

Project Goal

The primary goal of the project is to automate educational reporting processes within juvenile facilities by replacing manual spreadsheets with a secure digital platform. This will improve operational efficiency, data accuracy, compliance and strategic decision-making.

Project Objectives

- Digitalise Data Entry: Eliminate manual processes through structured online forms with validation rules.
- Automate Calculations: Compute results automatically to reduce calculation errors to zero and save teachers 60% of preparation time.
- Generate Report Cards: Produce customizable, print-ready and digital report cards.
- Provide historical tracking of progress, attendance and behaviour via student profiles.
- Embed Analytics: Give school leaders real-time visibility into performance summaries and statistics.
- Ensure POPIA Compliance: Secure data via role-based access, encryption and audit trails with zero critical findings.

Priority: Schedule: 1 Quality:2 Cost:3

Terms of References

- The project delivery deadline is set to September 2026.
- The system will be built using ASP.NET Core (Blazor Pages), SQL Server and Microsoft Azure.
- No system development can start without explicitly receiving Western Cape Government dev/staging environment access and formal sign-off.
- The team rely on free, open-source tools, libraries and frameworks. There is zero budget for paid licenses or learning resources.
- The project must follow an agreed agile life cycle methodology where work is evaluated in increments at designated milestone review points.
- End-users (teachers and facility administrators) are required to participate in user acceptance testing and supply feedback within pre-agreed timelines.
- All project tasks must be centrally managed and tracked using a shared Trello task board.
- Internal expectations, assigned roles, and conflict resolution rules were finalized in a signed team charter upfront.

Business Deadline	**Budget**
End of September 2026	20 hours per week for 5 months

Assumptions

- Resource Delivery: The project sponsor will provide all necessary information and resources.
- Skill Development: The student team is prepared to learn the necessary technical skills.
- User Training: Facility staff are willing to undergo training to operate the system.

- Timely Approvals: Sponsors and stakeholders will grant all approvals on time.
- Project scope and system requirements will remain stable once development begins.
- Testing Collaboration: Teachers and facility administrators will actively participate in user acceptance testing.
- A single point of contact will be assigned for rapid decision-making.
- Internet Stability: A reliable internet connection will be consistently available at the facilities.
- Hosting Compliance: The hosting environment will fully comply with statutory data storage rules.

Related Projects

- INF3003W workshops using ASP.NET Core, Razor Pages/MVC features.

Moral/Ethical/Legal Issues

- Handling juvenile data demands strict compliance with the Protection of Personal Information Act (POPIA).
- Mishandling sensitive student profiles creates severe security, data breach, and compliance risks.
- Safeguards must be verified by mandating a Privacy Impact Assessment (PIA) before deployment.
- Weak role-based controls risk exposing sensitive youth records to unauthorized personnel.
- The system must log an immutable audit trail for every change to maintain data integrity.
- Future shifts in the Children's Act or Child Justice Act may mandate rapid system reconfigurations

Risks Assessment

In this section, key risks that are both associated with the development of the ERPS and in scope with Team 8's involvement with the project are identified and analysed. Each risk is appraised and ranked by the intersection of their impact on the project's development and the likelihood of occurrence. Additionally, each risk identified is accompanied by a mitigation strategy within the Risk Response Plan.

Risk Matrix:

Overall threat level Key:

Green: Low Risk	Orange: Medium Risk	Red: High Risk

		Probability		
		High	**Medium**	**Low**
Impact	**Major**		R1; R12; R2; R3; R4	R7; R8; R6; R13
	Moderate	R11, R10, R9	R16; R5; R17	R18
	Minor	R14		R15

Risk Status Key:

Colour	RAG status	Action
Red	We have passed the trigger, and the team/project is now experiencing the risk. The project is likely to be negatively impacted by the risk.	The risk owner needs to immediately activate the contingency plan and notify all relevant stakeholders.
Amber	We are very close to realizing/experiencing the risk. We are close to the trigger and if nothing happens (or it's just a matter of time) we will be exposed to the risk.	Risk owner to ensure that mitigation plans in place have been activated to reduce the chances of the risk happening. Can proactively notify stakeholders likely to be affected should the risk be realized.
Green	The project is performing to plan. The specific risk hasn't been realised and there's no indication it's about to be realised.	No action is needed.

Risk Response Plan:

No	Rank	Risk	Description	Category	Cause	Triggers	Owner	Mitigation	Contingency	RAG Status
R1	1	Inadequate risk management	Poor risk identification and analysis. No mitigation strategies are considered.	Project	Failure to identify and monitor project risks	Risks are not being reviewed for more than 30 days, or more than 5 high-priority risks remain without mitigation plans	Project Manager	Develop a risk register, with mitigation strategies included. Risks mentioned should include those found in research and from possible past risk registers of similar projects. Review and update the register frequently.	Escalate significant risks to the project sponsor and update the risk management plan immediately	Green
R12	2	Poor requirement gathering	Requirements gathered are	Project	Insufficient stakeholder consultation	Development begins before 100%	Business Analyst	Have meetings with the stakeholders,	Conduct additional requirement	Green

			incomplete and incorrect.			stakeholder approval is obtained or more than 20% of requirements change after approval		document requirements and obtain final sign-off from stakeholders before project development can begin.	gathering sessions and update the documentation before development continues.	
R2	3	Incompatible system	Fails to integrate with existing systems and structures.	Integration	Existing systems use different technologies or data formats	More than 10% of integration tests fail or any critical external system refuses data exchange during testing	Project Team	Conduct early analysis on all integration points, which will inform the system requirements. Conduct integration testing before final deployment.		Amber
R3	4	Poor or substandard design	The design creates functional, security, or performance inefficiencies and failures.	System	Insufficient design review or lack of user feedback	Design approval is obtained from less than 80% of representative users	Project Team	Construct a detailed quality assurance program that must include testing done by end users.	Redesign affected components and repeat testing before release.	Green

R4	5	Sudden power outages	Loss of power means the system will shut down. If this is sudden, it is unexpected, meaning any manual saves might not have been completed, and data may be lost.	Environment	Load shedding or electrical failures	A load shedding event or unexpected power failure exceeding an hour occurs within a month	Project Sponsor	Include autosave functionality in the system.	Restore latest saved version and ensure that lost data is captured again.	Green
R11	6	Scope Creep	additional features and requirements are continuously added without regard for resource, budget, or time constraints.	Project	Uncontrolled addition of new requirements	Scope increases by more than 10% of the approved requirements or more than 3 change requests are submitted in a month after scope sign off	Project Manager	Define and document the scope through a scope statement, work breakdown structure, and product breakdown structure. Implement a formalised change control process that requires sign-off by the	Review initial scope, prioritise tasks based on impact, and postpone additional requirements	Amber

								project sponsor.		
R10	7	Loss of documentation	The only copy of documents (such as scope agreement, team charter, task allocation, and contribution tracing worksheets) are lost, stolen, or destroyed.	Project	Files accidentally deleted or storage failure	Backup verification fails during monthly audit.	Project Manager	Store documentation digitally in a version-controlled repository. Conduct regular audits of the repository.	Recover documents from backups or version history, if not available redo documentation	Green
R9	8	Unplanned team member unavailability	Team members are briefly yet suddenly unavailable, meaning they cannot compete their allocated tasks or contribute to the project development	People	Illness, emergencies or personal commitments	A key team member becomes unavailable for more than 3 consecutive days during a critical project activity or task completion falls below 80% of	Project Manager	Document all assigned tasks and current contributions to streamline emergency reallocation of tasks. All current knowledge should also be documented to prevent misinformation. Each task should be	Reassign tasks to available team members and adjust the schedule if necessary	Green

						schedule expectations.		assigned a backup executor, and thus, staff should be multi-skilled and up to date.		
R16	9	Missed deadlines	Tasks are not completed by the associated deadline.	Project	Poor estimation, delays, or dependencies	More than 20% of critical tasks are delayed by more than 3 working days, causing dependent tasks to start late.	Project Team	Construct a detailed project schedule, accounting for all deadlines and slight buffer times to critical tasks. Utilise a network diagram or a Gantt chart.	Re-prioritise tasks, allocate additional resources, and revise the project schedule.	Green
R5	10	Poor or non-existent internet connectivity	If the system is configured to require an internet connection, the lack thereof will render it unusable.	Environment	Network outages, unstable internet, and high data costs	Network availability drops below 95% uptime in a month.	Project Sponsor	Design the system with offline capabilities and develop performance while accounting for various network	Use offline mode and synchronise data once connectivity returns	Green

								connectivity scenarios.		
R17	11	Weak sponsor engagement	Difficulty communicating, acquiring assistance, etc.	Adoption	Sponsor unavailable or slow to respond	Sponsor fails to respond to approval or feedback requests within 5 business days or misses 2 consecutive meetings.	PM	Establish communication channels for emergency escalation, such as alternative means of contact.	Find an effective communication channel that works for the sponsor or escalate urgent decisions through alternative contacts and continue work on approved tasks	Green
R7	12	Weak role-based access control	What should be locked behind certain ranks or responsibilities are not configured to express that.	System	Incorrect user permissions or poor security configuration	Security audit identifies 1 or more unauthorized access permissions	Project Team	Define all role-specific workflows and set user access and authorisation to reflect that documentation.	Remove incorrect permissions immediately and conduct a security audit	Green
R8	13	Inadequate privacy protection measures	Gaps in the system's privacy assurance during data	Legal	Failure to implement adequate security controls	Sensitive learner information is accessed, or a privacy	Project Team	Mandate the completion of a Privacy Impact Assessment	Contain the breach, notify management, investigate the investigate the	Green

			collection, analysis, and storage.			audit records one or more non-compliance findings.		(PIA) before system deployment.	incident and implement corrective measures	
R6	14	Poor data quality	i.e., incomplete records, inconsistent data entry and storage, errors with manual inputs	System	Human error during data entry or incomplete records	More than 5% of records contain missing or invalid data or data validation errors exceed 3% per month.	Project Team	Ensure data verifications and validation controls.	Perform data cleansing and request users to recapture missing information	Green
R13	15	Poor team management	the team is in constant conflict which is not resolved. Additionally, the team has limited cohesion and direction.	People	Lack of communication or unresolved conflict	More than 2 unresolved team conflicts remain open for over 2 weeks.	Project Team	A clear project manager should be appointed to the development team. Set up a project charter wherein all team roles, expectations, conflict resolution processes, communicatio	Escalate unresolved conflicts to the project manager and redistribute work if necessary	Green

								n, and the like are clearly defined. All team members should sign off after being given time to understand and potentially negotiate the terms. Regular team and individual check-ups should be conducted.		
R14	16	Lack of resources	allocated resources are below the amount needed to complete the project.	Resources	Insufficient staff, equipment, or software	More than 30% of required hardware, software, or personnel are unavailable for more than 5 working days.	Project Sponsor	During planning, conduct resource requirement analysis and implement contingency plans for resource procurement.	Request additional resources or reduce project scope to fit available resources.	Green

R18	17	Legislation changes	Major changes in relevant legislation (i.e., POPIA, Children's Act, Child Justice Act) that will require unplanned changes in the project to maintain compliance.	Legal	Amendments to existing legislations.	New legislation affecting the system comes into effect with a compliance deadline within 12 months.	Project Team	Conduct regular research and analysis on relevant legislation to ensure continued compliance. Design the system with configurable business rules to ensure quick legal compliance capability	Review compliance requirements, update system specifications, and implement the required changes before deployment.	Green
R15	18	Budget constraints	available funds are below the amount needed to complete the project.	Project	Costs exceed the approved budget	Project spending reaches 90% of the approved budget before 80% of project completion.	Project Sponsor	Construct a detailed budget with contingency reserves. Monitor expenses and conduct regular budget reviews.	Reduce non-essential features, seek additional funding, or extend the project timeline if approved.	Green

Table 2: Risk Response Plan

Business Processes

This section makes use of an activity diagram, a type of Unified Modelling Language (UML) figure, to describe the sequence of actions carried out by several actors within the system to complete an activity. This type of diagram excels at visualising workflows that involve many actors and decision points, especially when comparing the complexity of current and proposed systems.

The activity diagram below depicts the proposed system to be used by the DSD, whereby educators capture and report on the academic performance of students within juvenile facilities. The process begins after assessments have been conducted and collected. Educators log into the system and are presented with a list of subjects for which they have marking authority. They then select the relevant subject and are shown the assessments available for marking.

After selecting an assessment, the educator is presented with a list of students registered for that assessment. Selecting a student opens the mark-entry interface, which may include either a rubric-based marking scheme or a field for direct mark entry. The educator enters the learner's mark, and the system performs validation checks to ensure that the mark is entered in the correct format, that the value is numeric, conforms to permitted decimal precision, and does not fall outside of the assessment's minimum or maximum allowable marks. If the entered value is invalid, the educator is prompted to correct and resubmit the mark. Otherwise, the mark is saved to the database, and the educator may proceed to the next student.

Once the marking process has been completed and the draft period has passed, educators' ability to create, modify, or delete marks is restricted. At this stage, the facility or education manager may request the generation of academic reports for the facility. The system accomplishes this by retrieving the latest mark data for each student from the database, performing the necessary calculations (including term averages, subject totals, pass or fail status, credits acquired versus credits required, and promotion status), and subsequently populating the report template. Note that the calculated results produced during this process are also stored in the database to support year-level statistics and facilitate further analytical processes.

The facility manager then reviews the generated reports and, if errors are identified, sends amendment requests to the appropriate educator, granting temporary editing access. Once the amendments have been completed, the reports are regenerated and reviewed again. If no issues are found, the reports are approved and returned to the facility manager, who may then distribute the final report cards to the relevant parties.

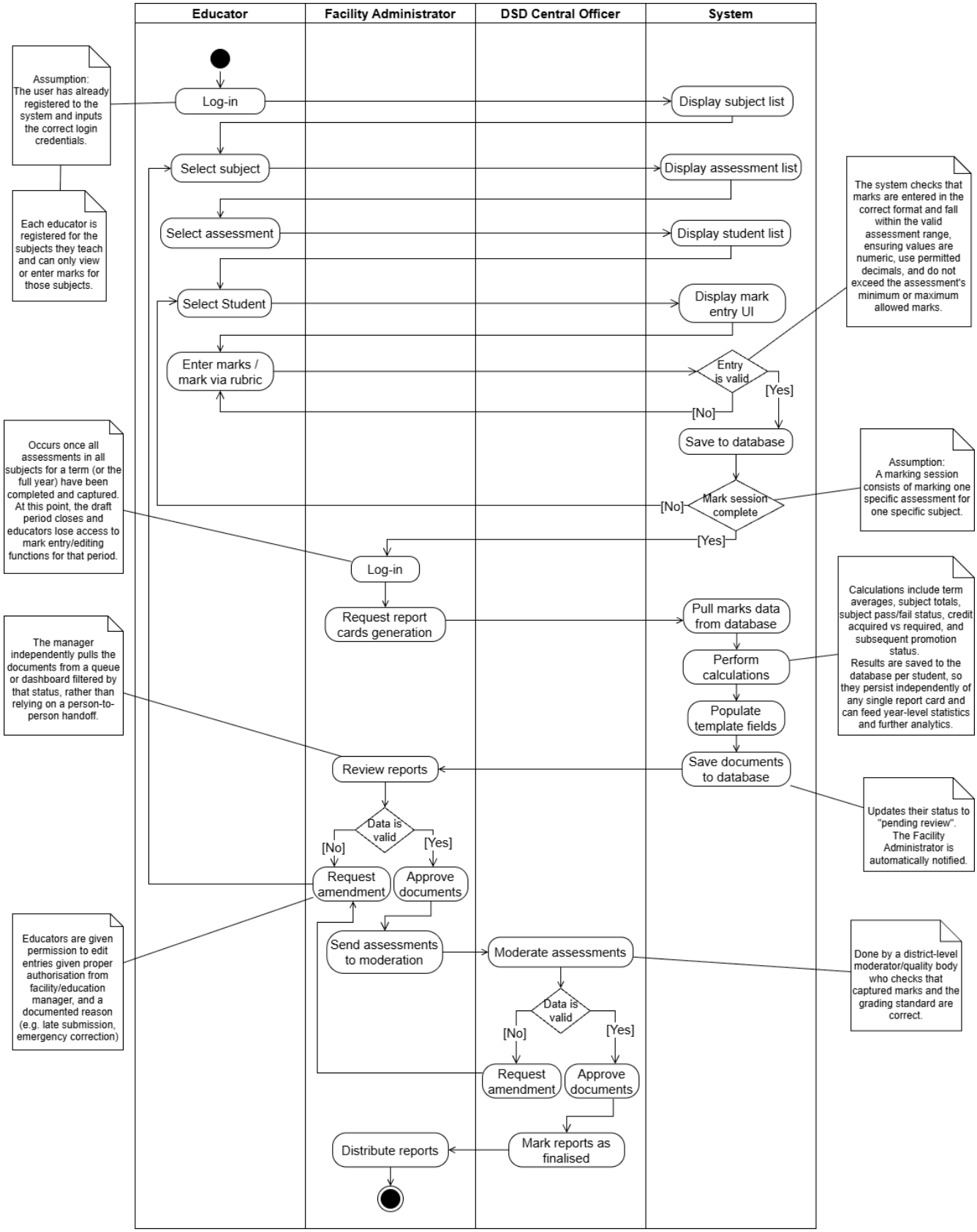


Figure 1: To-Be Activity Diagram depicting the mark entry and report generation processes

System Scope

A system scope is the boundaries of the system in terms of what it will and will not do. For EPRS specifically, it means clearly stating that the system will handle learner data capture, report card generation, analytics, and integrations with DHET and DSD. A medium-level package diagram aids to visually explain where the system starts and where it will end.

Medium Level Package Diagram

A package diagram is a visual map of how a system is organised into different logical groups based on functionality. Each package is a named box representing a group of related features or components and arrows which show dependencies between packages. The main purpose of a package diagram is to show the thought process of what the system is able to achieve. A medium-level package breaks a high-level package into more specific functional areas. It provides more detail but still does not show individual use cases.

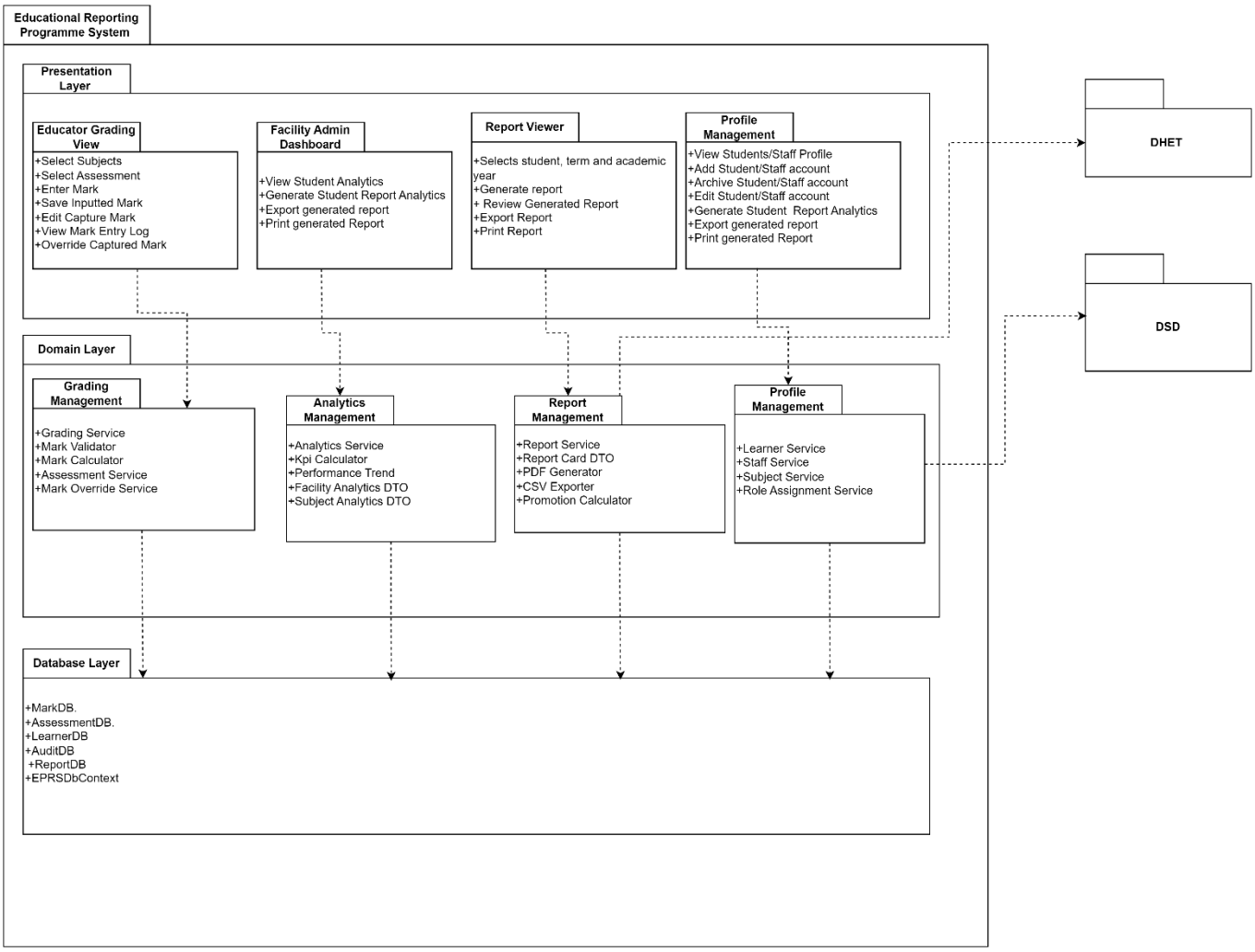


Figure 2: Medium-Level Package

The presentation package serves as the user interface of the system. It provides tailored views for the different user roles, namely the Educator Grading View, Faculty Administrator Dashboard, Report View, and Student/Staff Profile View. Through these interfaces, users access the system's core functionality without interacting directly with the underlying data storage.

The Grading Management package is responsible for all assessment-related activities. Educators can select subjects and assessments, enter and edit learner marks, save captured marks, view mark entry logs, and override captured marks where authorised.

The Analytics Management package provides reporting and analytical capabilities. It retrieves assessment and learner information from the database to generate student performance analytics and reports. Users can view student analytics, generate reports, and export or print these reports for academic and administrative purposes.

The Report Management package is responsible for producing official reports required by the institution and external stakeholders. Faculty administrators can select the relevant student, academic term, and academic year before generating reports. Reports may then be reviewed, exported, or printed. This package communicates with the DHET external system to submit the required institutional reports.

The Profile Management package manages learner and staff information. It allows authorised users to view profiles, add new student or staff accounts, edit account information, archive inactive accounts, and generate profile-related analytics. The package depends on the DBMS to retrieve and update user information. It also provides reporting outputs that may be shared with the DSD where required.

At the centre of the architecture is the Database Management System (DBMS) package, which acts as the system's central data repository. It is divided into four logical storage components. The Student Mark Storage stores learner assessment marks, Learner and Staff Database stores user profiles and account information, Document Storage stores generated reports and supporting documents, Audit Log Database, which records significant system events for accountability, monitoring, and auditing purposes.

Finally, the system integrates with two external organisations. The Department of Higher Education and Training (DHET) receives official educational reports generated by the Report Management package, while the Department of Social Development (DSD) receives profile and programme information generated through the Profile Management package. These external dependencies enable compliance with reporting and regulatory requirements.

Overall, the separation of responsibilities improves maintainability, scalability, and security by ensuring that presentation, business logic, data management, and external integrations remain loosely coupled. Together, these packages enable the ERPS to efficiently manage learner information, assessments, reporting, and regulatory compliance.

Class Diagram

An analysis class diagram is a visual map of how data and responsibilities are structured within the system. Each box represents a specific entity or user role containing its own attributes and methods, while the connecting lines and arrows show how these components relate to one another. The main purpose of an analysis class diagram is to show the logical thought process behind what information the system stores and how different data objects interact to support the business processes.

This system captures students marks and tracks student academic performance within a Juvenile Facility. A Facility Administrator oversees the facility and compiles formal Report Cards for the registered Students. Students register for learning area (subject) through Enrolment class to resolve many-to-many relationship. On the curriculum side, an Educator teaches one Subject, sets specific Assessments for them, and records the results into a CapturedMark entry. Meanwhile, an Education Manager reviews these

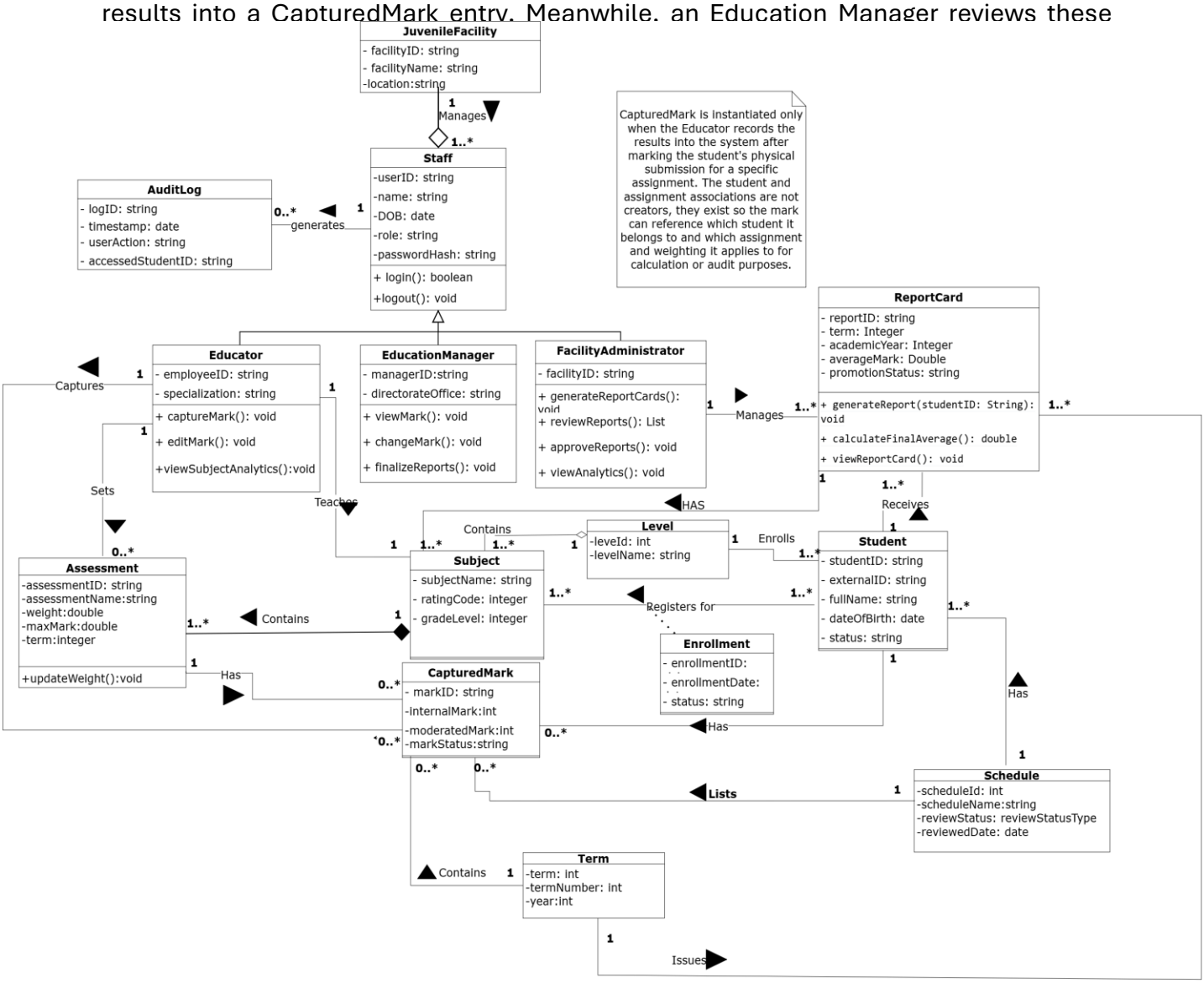


Figure 3: Class Diagram

Proposed Functionality

Use Case Diagrams

A use case diagram is a visual, high-level representation of how users interact with a system to achieve specific goals. It acts like a behavioural map, outlining who is using the system (the actors) and what they are doing (the use cases). It is used to specify the details of the system, capture the requirements of the system and gather an understanding of how the user works with the system.

Authorization Use Case diagram

The Authorization diagram shows how facility administrators interact with the security and access control layer of EPRS. The user must first pass through the login with credentials use case before they can access any other part of the system. Once authenticated, the system verifies the user's role permissions to determine what they are allowed to see and do. The system administrator is the only actor who can manage user accounts and assign roles, with both of these use cases to ensure all access decisions are compliant.

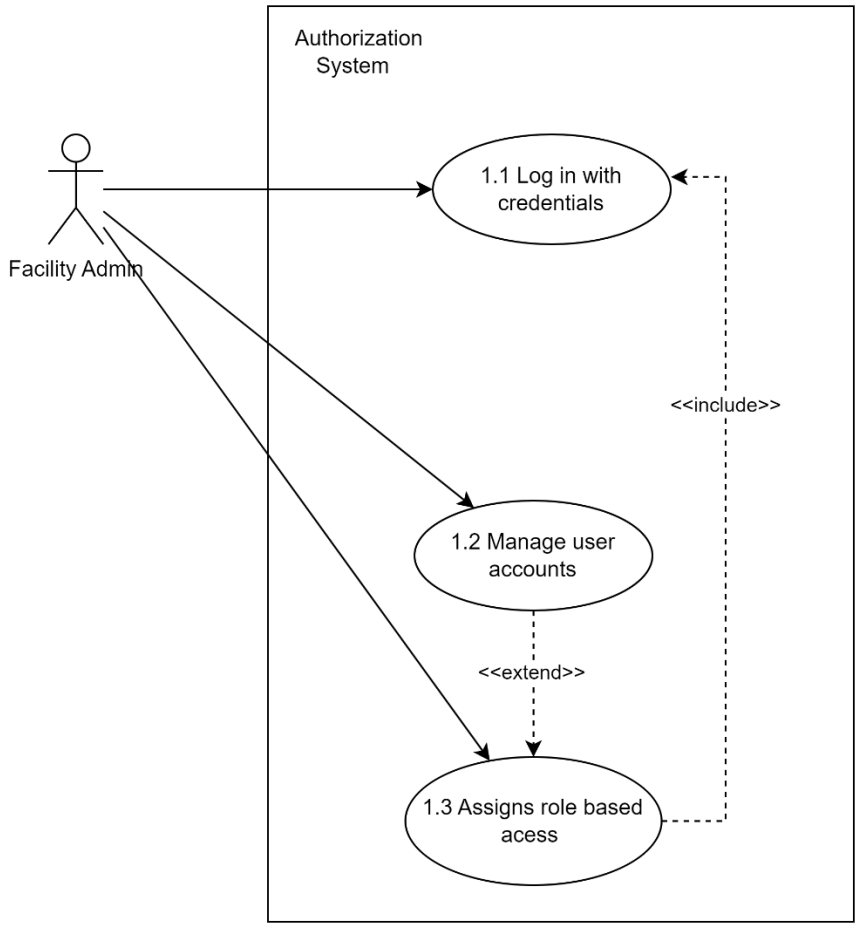


Figure 4: Authorization Use Case Diagram

Mark Capture System Use Case diagram

The Mark Capture System diagram focuses on the day-to-day data capture workflow. The educator is one of the primary actors and can browse subjects, view the assignment list, capture marks, and edit existing marks while the other primary actor, educator administrator, has the capability to view the mark entry log to track when marks have been entered into the system. The diagram shows that browsing subjects and viewing the assignment list are sequential steps that must happen before any mark can be captured, represented by the include relationship between them. Editing an existing mark extends the capture use case, meaning it is only triggered when a mark already exists for the selected learner. This subsystem is the core of EPRS and directly replaces the current manual Excel and Word process.

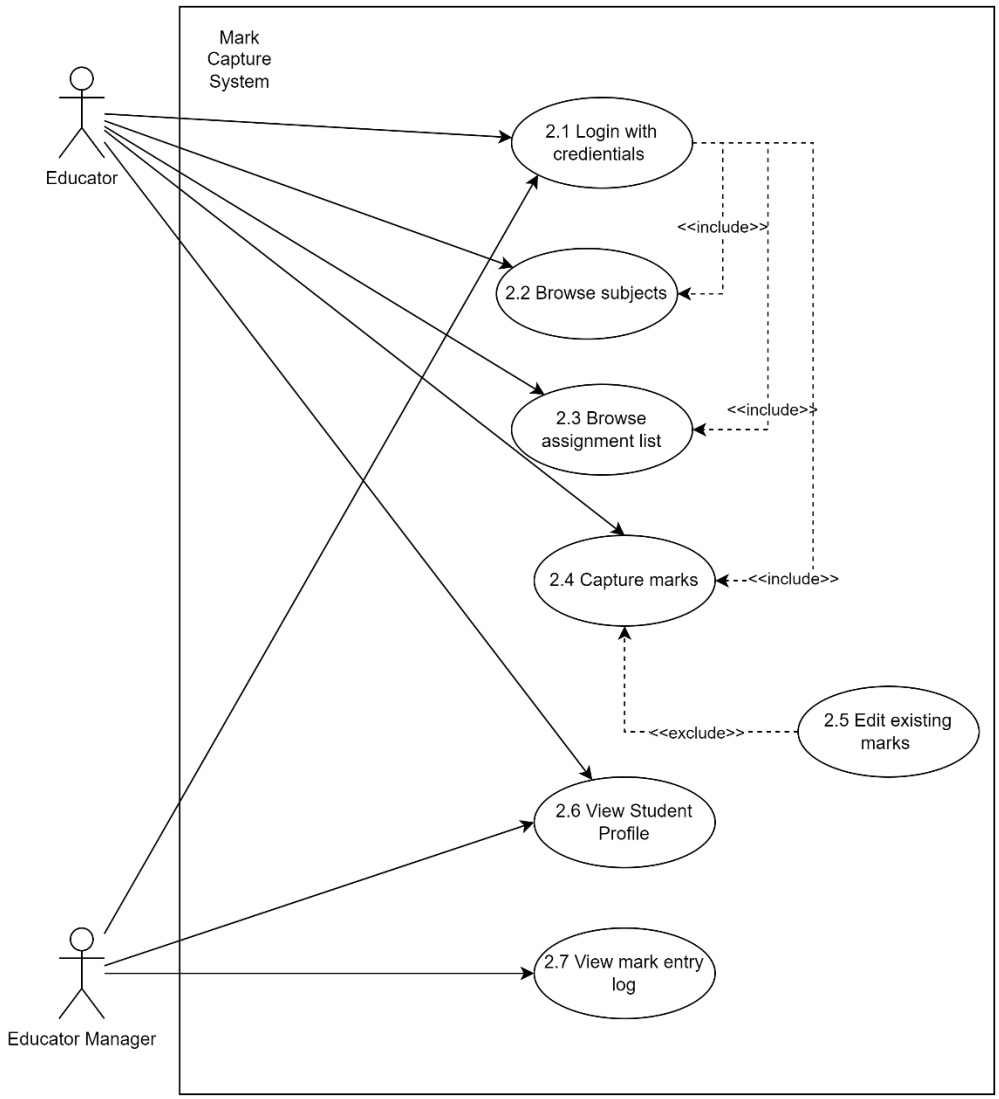


Figure 5: Mark Capture Use Case

Report Generation Use Case Diagram

The Report Generation subsystem diagram shows how official academic student reports are produced based on the marks captured by educators. The facility administrator initiates report generation by selecting a specific level, term, and academic year. An official academic report card is then generated containing the student's personal details, individual subject marks, overall average, and promotion outcome. The export report use case extends report generation, allowing the completed report card to be downloaded as a PDF for physical filing or digital sharing. The print report use case similarly extends it, sending the formatted report card directly to a printer for hard copy distribution to the facility or the student's file. This subsystem is one of the most valuable parts of EPRS because it eliminates the manual process of compiling student academic reports from multiple spreadsheets.

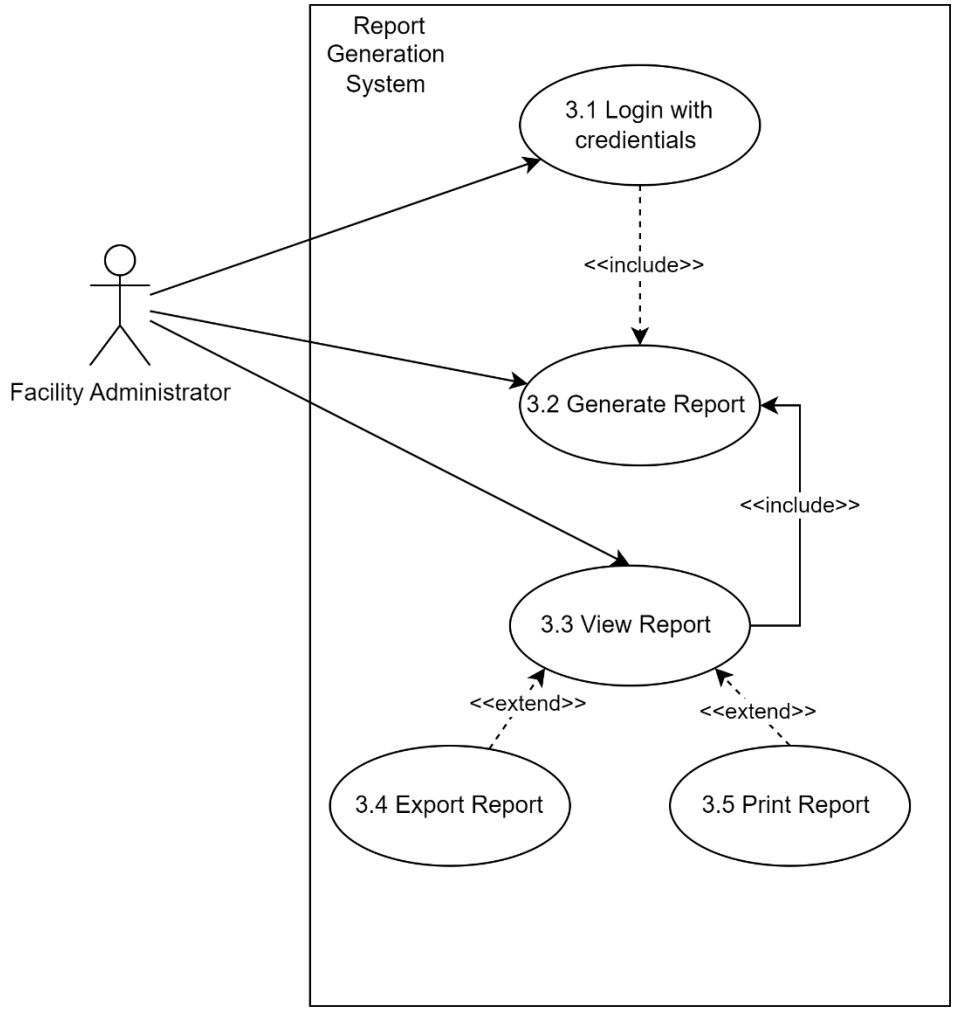


Figure 6: Report Generation Use Case

Dashboard Analytics Use Case diagram

The Dashboard Analytics Use Case diagram shows the reporting and oversight functions available to facility administrators, educator and educator manager. The educator and educator manager can view student-level KPIs. The facility administrator operates at a higher level and can view facility analytics, generate official reports, and access the audit log. Report generation is the central use case in this diagram, with export report and print report both extending it which means they are only triggered after a report has already been generated. This subsystem directly addresses the strategic planning gap identified in the problem statement by giving WCG management real-time visibility into educational performance across the entire facility

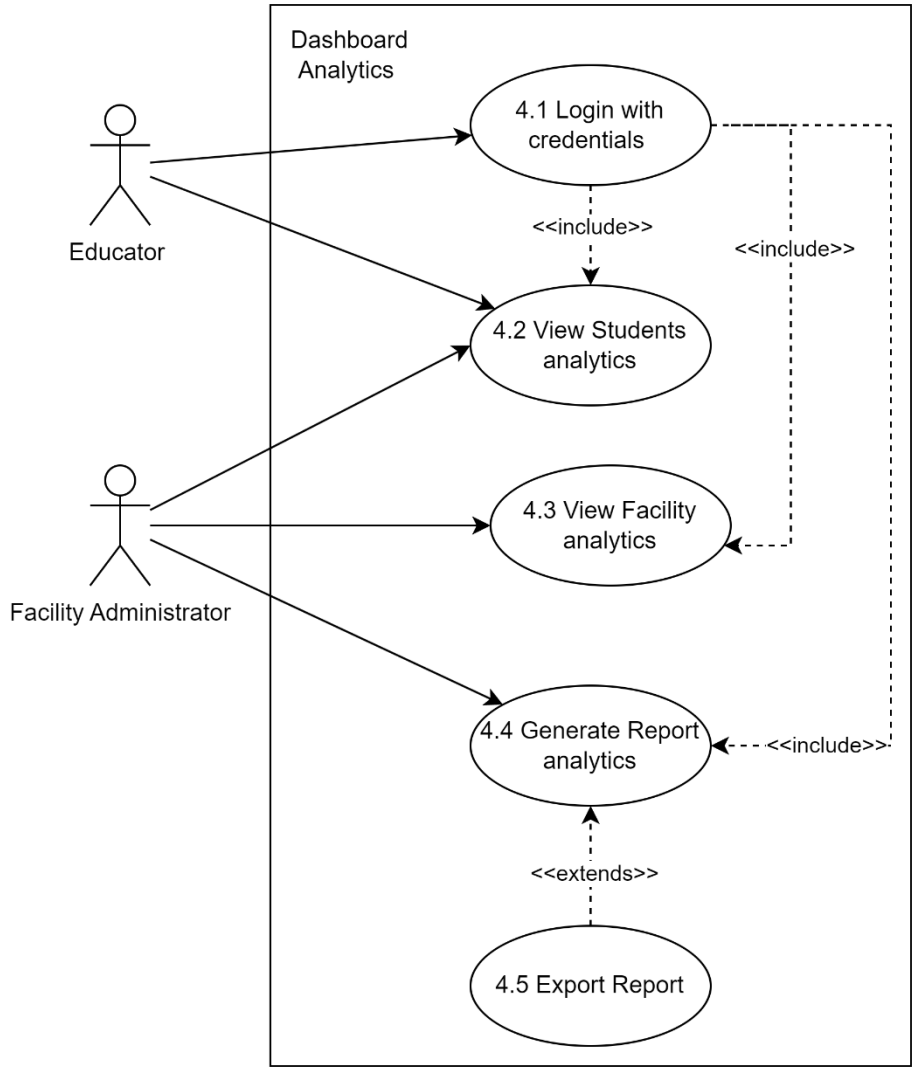


Figure 7: Dashboard Analytics Use Case

Use Case Narratives

Use Case Name: Manage Access Control	**ID**:1.3	**Importance level**: High
Primary Actor	Facility Administrator	
Secondary Actor	None	
Stakeholder and Interest	Western Cape Government (POPIA-compliant access control) WCG CeI (secure, auditable account management) Learners (protection of their personal information from unauthorised access)	
Brief description	Allows System IT to assign appropriate roles within EPRS.	
Preconditions	The system administrator is logged in with system administrator privileges.	
Postconditions	The user account is created, updated, or deactivated with the correct role-based permissions applied. The changes are reflected on to the database	
Related Use Cases	Include: 1.1 Login with credentials	
Flow of events	**Actor**	**System**
	1. Selects a user account on the ERPS	2. Display the account details
	3. Selects the appropriate role whether Educator, Facility Manager or System IT	4. Validates the role that is appropriate for the user's facility and position
	5. Confirms the role assignment	6. Applies the role-based access control restrictions
Alternative flow of events	3a. Conflicting role already exists for the user	3.1 The system displays a warning that requires the existing role to be removed before a new one is assigned
	3.2 Removes the existing role that the user is assigned	3.3 Returns the process to Step 3 (Select the appropriate role) so the administrator can assign the new role.

Table 3: Manage Access Control Use Case

Use Case Name: Capture Students Marks	**ID:** 2.4	**Importance level:** High
Primary Actor	Educator	
Secondary Actor	None	
Stakeholder and Interest	Western Cape Government (accurate learner records for compliance reporting) Learner (fair and correctly recorded academic progress) Educator (reduced administrative burden compared to manual capture)	
Brief description	Allows an educator to record a learner's mark for a specific assignment within a selected subject, replacing the manual Excel and Word process.	
Preconditions	Educator is logged in. A subject and assignment have been selected. The learner is enrolled and active.	
Postconditions	The mark is saved to the database and reflected immediately in the learner register with a captured status.	
Related Use Cases	Include: 2.2 Select subject Include: 2.4 Select assignment Extend: 2.5 Edit captured mark	
Flow of events	**Actor**	**System**
	1. Navigates to list of subjects	2. Retrieves and displays all the subjects that the user teaches
	3. Selects a subject <<include 2.3 Select Subject Use Case>>	4. Retrieves and displays all the assignments that are under the subject
	5. Selects a specific assignment <<include 2.4 Select Assignment Use Case>>	6. Retrieves the list of students under the assignment
	7. Select a learner from the learner register	8. Displays the mark entry form which is pre-populated with learner and assignment details
	9. Selects whether the student has completed the assignment, handed in late or copied from another student using a checkbox	

	10. Enter marks that the student achieved	11. Validate whether the mark falls between 0 and 100, Validate that the input is a number and not letter
	12. Optionally add comment	
	13. Saves the mark	14. Stores the mark in the data base
		15. Writes and saves the captured mark in the audit log
	16. Clicks the "next student button"	17. Displays the next mark entry form for a new student
Alternative flow of events	10a. User input an invalid mark	10.1 Displays an error message to user
		10.2 Prevents user from saving the mark until error is resolved
	9a. User selects that the assignment is incomplete	9.1 Disables the mark entry input box
	9b. User selects that the assignment was handed in late	9.1.1 Displays and enables a text box that lets the user enter how late the assignment was
	9.1.2 Enters how late the assignment was	9.1.3 Calculates the mark the student will get with a late submission penalty

Table 4: Capture Student Marks Use Case

Use Case Name: Edit Mark Captured	**ID:** 2.5	**Importance level:** High
Primary Actor	Educator	
Secondary Actor	Educator Manager	
Stakeholder and Interest	Western Cape Government (accurate, correctable records for compliance) Learner (correct reflection of their actual performance)	
Brief description	Allows an educator to correct or update a mark that has already been captured for a learner and assignment.	
Preconditions	A mark already exists for the selected learner and assignment Editing of a mark can only be done during a designated term period	
Postconditions	The learner's mark reflects the corrected value. The original entry is overwritten rather than duplicated	
Related Use Cases	Extend: 2.4 Capture student mark	
Flow of events	**Actor**	**System**
	1. Select a learner whose marks has already been captures	2. Loads the mark entry form with the existing marks and comments
	3. Updates marks or comments	4. Validates the updated mark falls within range of 0 and 100
	5. Saves changes	6. Updates the existing record in the database
		7. Writes and saves the edit in the audit log
Alternative flow of events	1a. User input an invalid mark	1.1 Displays an error message to user
		1.2 Retains the previous saved value until a valid mark is entered

Table 5: Edit Mark Captured Use Case

Use Case Name: View Mark entry log	**ID**: 2.7	**Importance level**: Medium
Primary Actor	Educator Manager	
Secondary Actor	Educator	
Stakeholder and Interest	Western Cape Government (accountability and traceability for POPIA compliance) WCG CeI (ability to investigate discrepancies in learner records)	
Brief description	Displays a chronological record of mark entries and edits for audit and oversight purposes.	
Preconditions	At least one mark has been captured or edited in the system.	
Postconditions	The user has visibility of who captured or edited each mark, and when.	
Related Use Cases	Extend: 2.4 Capture student mark Extend: 2.5 Edit captured mark	
Flow of events	Actor	System
	1. Navigate to the mark entry log	2. Retrieve the audit log filtered by educators or date range
	3. Search or filter the mark entry page	4. Displays who captured or edited each mark and when
Alternative flow of events	3a. No entry matches the selected filter	3.1 System displays an empty state message

Table 6: View Mark Entry Log Use Case

Use Case Name: Generate Reports	**ID**: 3.2	**Importance level**: High
Primary Actor	Facility Administrator	
Secondary Actor		
Stakeholder and Interest	DHET (accurate compliance submissions) WCG management (reliable data for strategic planning) Facility (formal recognition of learner progress) Learner (accurate and official marks and progress)	
Brief description	Allows a Facility Administrator to generate an official report card for a learner based on their captured assessments for a given term and year.	
Preconditions	Learner must have completed the required assignments from a single term and year	
Postconditions	A report card is created and stored in the document store Report card is available for export or printing.	
Related Use Cases	Include: 3.3 View Report	
Flow of events	**Actor**	**System**
	1. Navigates to report generation	2. Displays the report generation form with fails for learner, term and academic year
	3. Select a learner, term, and year	4. Checks that at least one assessment has been captured for the selected learner within the selected term and year
	5. Confirm report generation	6. Retrieves all assessments for that learner within the specified term
		7. Calculates the average mark across all retrieved assessments
		8. Determine the promotional outcome, pass if the average meets or exceeds 30% and

		fail if the average is below 40%
		9. Generated the report card document, stored it in the database and creates an entry in the audit log
		10. Displays the completed report card with learner details, subject marks, average and promotion outcome
	11. Reviews the generated report	
	12. Selects export or print as required	13. Convert the report to either a PDF or CSV file and allows the user to download the generated report
Alternative flow of events	1a. No assessments exist for the learner in that term	1.1 Prevents report generation
		1.2 Notifies the user of the error and that there is no data available for the selected period
		13a. fails to export the generated report
		13.1 The system alerts the user of the error with and error message that allows the user to retry the process
	13.2 The manager can retry the process without losing the generated report	

Table 7: Generate Report Use Case

Use Case Name: View Facility Analytics	**ID**: 4.3	**Importance level**: High
Primary Actor	Facility Administrator	
Secondary Actor		
Stakeholder and Interest	Facility administrator (visibility into learner performance at their facility)	
Brief description	Allows a facility administrator to view aggregated performance data for learners at their facility.	
Preconditions	The facility administrator is logged in. Assessment data exists for at least one subject and assignment at the facility.	
Postconditions	The administrator has an accurate, real-time view of educational performance at their facility.	
Related Use Cases		
Flow of events	**Actor**	**System**
	1. Navigate to the analytics dashboard	2. Calculate average marks, pass rates, and subject-level breakdowns for the facility
		3. Displays the information using charts and KPI summaries
	4. User filters and adjusts by date, subject and level to view detailed progress	5. System dynamically recalculates to deliver a personalized view of user's filtered data
		6. Complies the analytic data for the subject into a formatted report and stores it in the database
		7. Generates an entry to the audit log
Alternative flow of events		2a. There is no captured assessment for the student selected
		2.1 Displays the student is a list view with no data available message rather than excluding them from the view

		2b. No assessments have been captured for a specific subject
		2.2.1 Displays an empty analytics state with a message prompting the user to capture marks before analytics can be generated
		7a. fails to export the generated report
		7.1 The system alerts the user of the error with and error message that allows the user to retry the process
	7.2 The manager can retry the process without losing the generated report	

Table 8: View Facility Analytics Use Case

Sequence Diagrams:

This section utilizes design sequence diagrams, a UML behaviour diagram, to depict use case interactions between actors and the system, given a 3-tier architecture (user interface, business logic, and data storage layers) setup.

The sequence diagram in Figure 8 describes the interaction between an admin member and the system's interface, controllers, entities, and database interactions that enable the view analytics functionality.

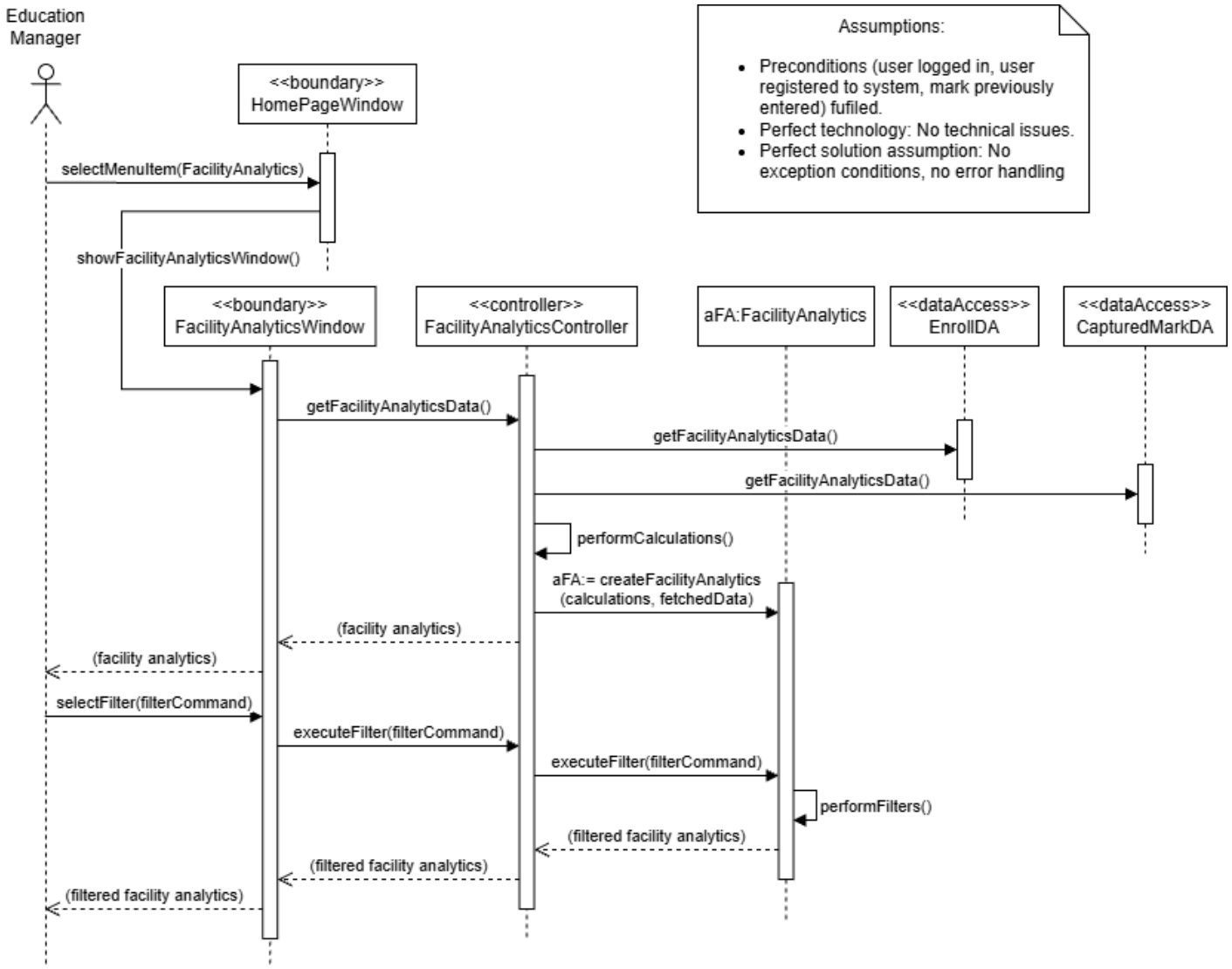


Figure 8: Sequence Diagram detailing View Facility Analytics use case

Figure 9Figure 8 describes the interaction between an educator and the system's interface, controllers, entities, and database interactions that enable the capture of marks.

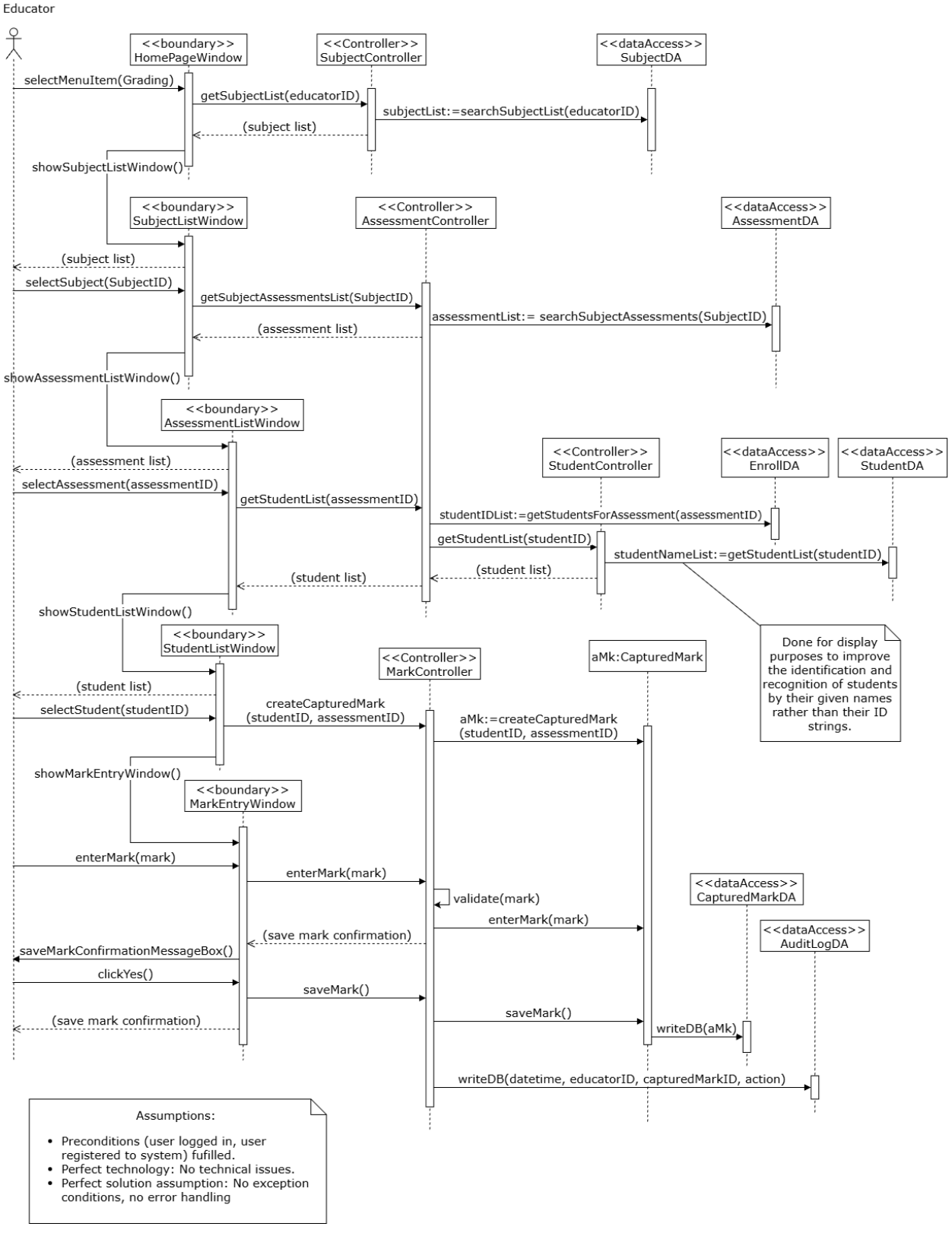


Figure 9: Sequence diagram depicting Capture Students Marks use case

Figure 10Figure 8 describes the interaction between an educator and the system's interface, controllers, entities, and database interactions that enable the alteration of captured marks.

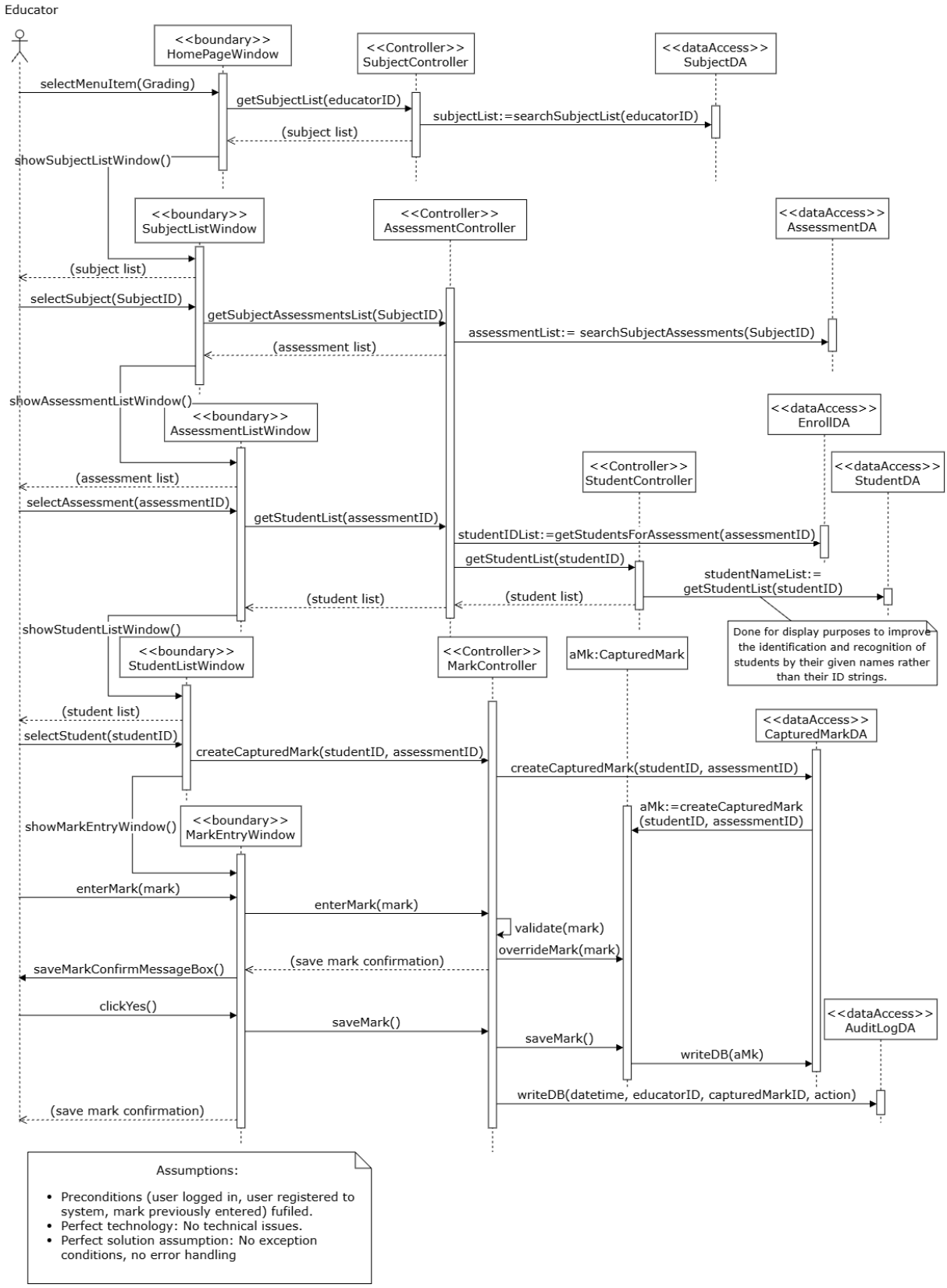


Figure 10: Sequence diagram depicting Edit Captured Mark use case

Figure 11Figure 8 describes the interaction between an educator and the system's interface, controllers, entities, and database interactions that enable access control.

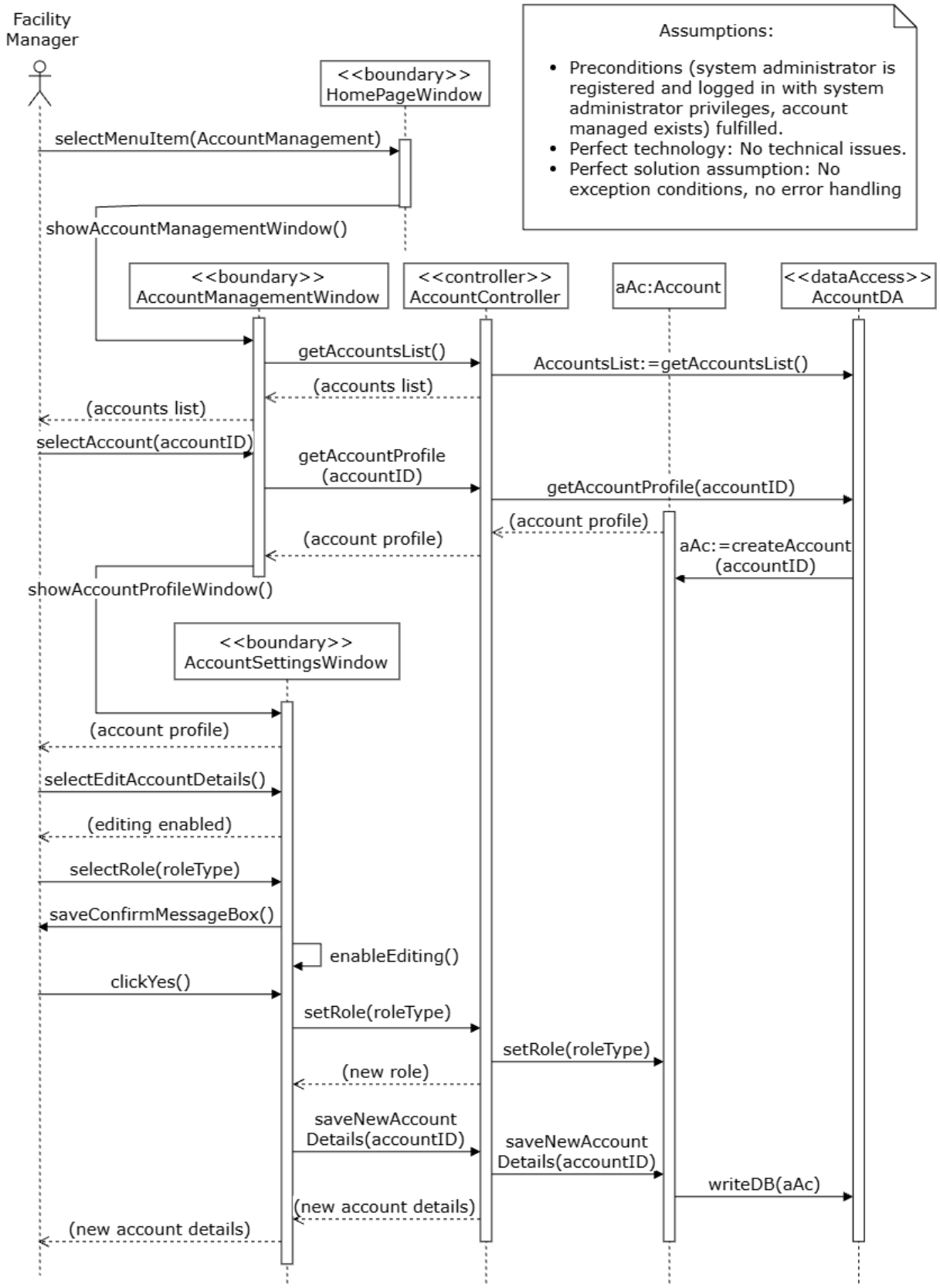


Figure 11: Sequence diagram depicting Access Control use case

Lastly, the sequence diagram in describes the interaction between an educator and the system's interface, controllers, entities, and database interactions that enable the report generation functionality.

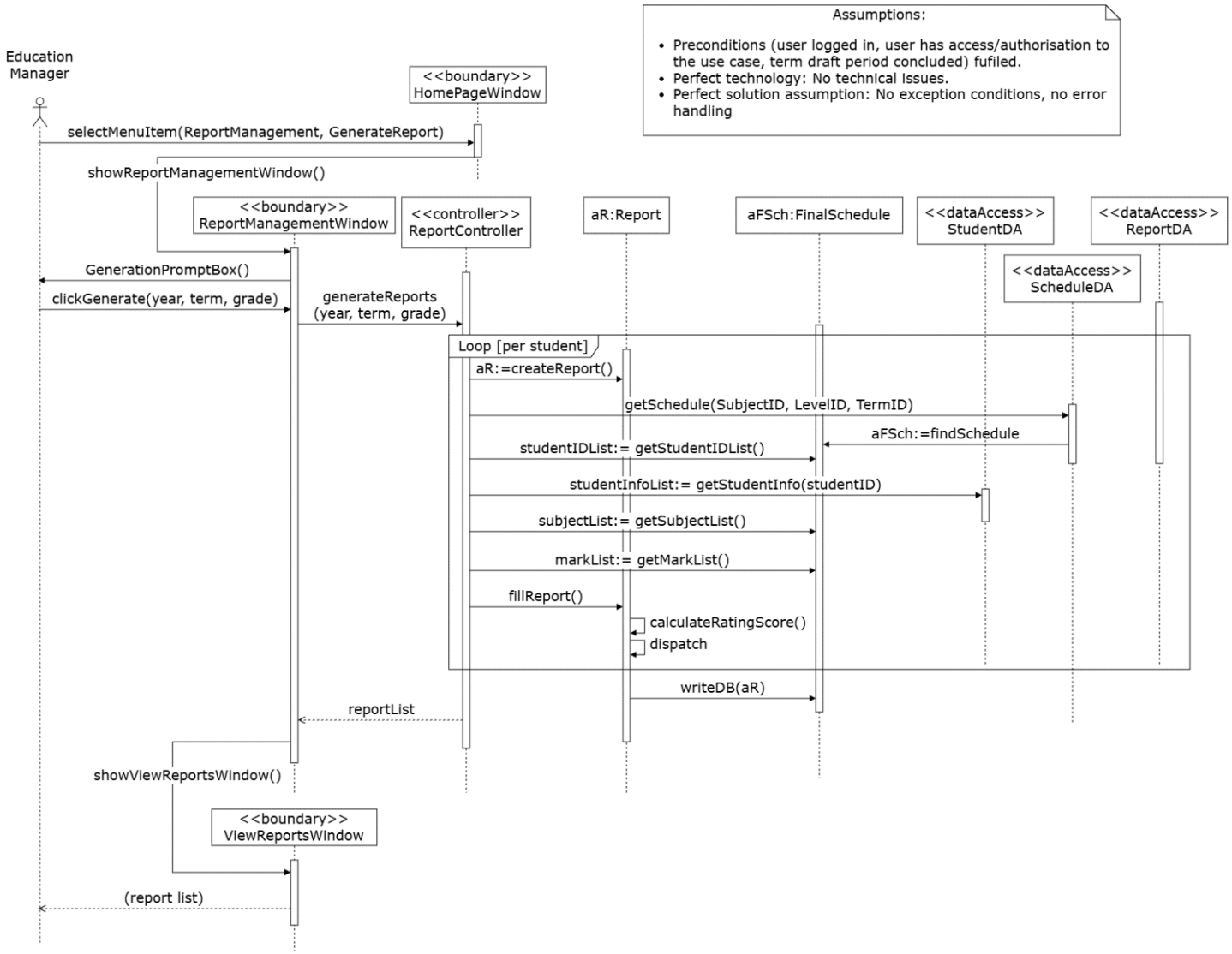


Figure 12: Sequence Diagram depicting Generate Report use case

State Machine Diagrams

A State Machine Diagram (SMD) is a UML behavioural diagram that depicts the various states an object within the system can undertake during its lifecycle and what. These states are determined by the values of the object's attributes and the relationship between objects. The transition between states is triggered by an object satisfying certain criteria, performing various actions, or waiting for events.

Figure 13 illustrates how a report card is generated, saved as a draft, and then queued for review within the database. Post-review, if it is rejected, it is sent back for amendment. Otherwise, it's accepted, finalised, and distributed through printing or exporting.

Figure 13: SMD depicting report cards

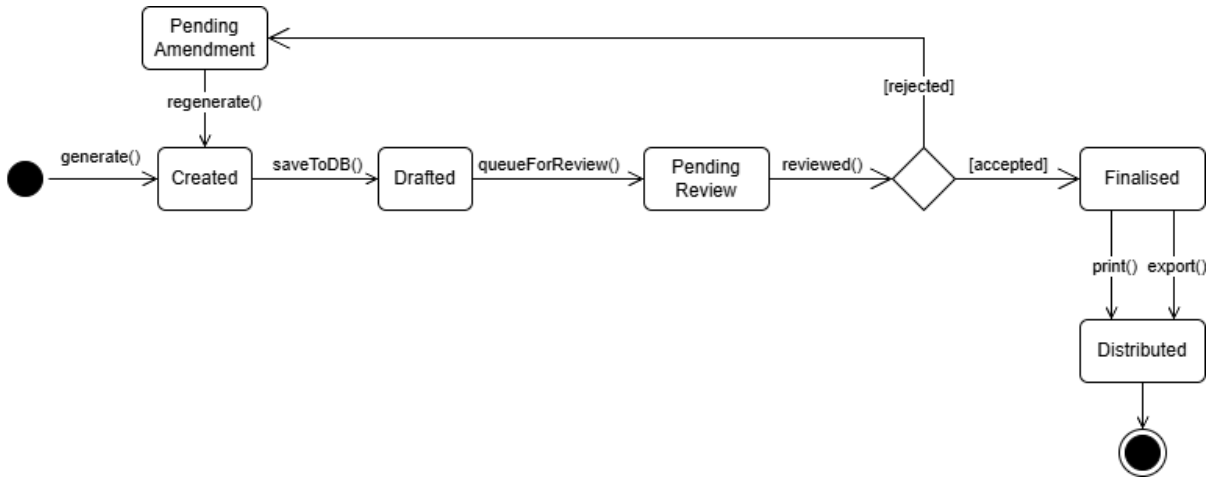


Figure 14 illustrates how a mark is entered and then validated. If rejected, it pends amendment. Otherwise, it is saved as a draft that can still be re-entered. After the draft period ends, the entry is frozen and becomes immutable while pending moderation. Upon acceptance, it's finalised; upon rejection, it is sent back for amendment.

Figure 14: SMD depicting mark entries

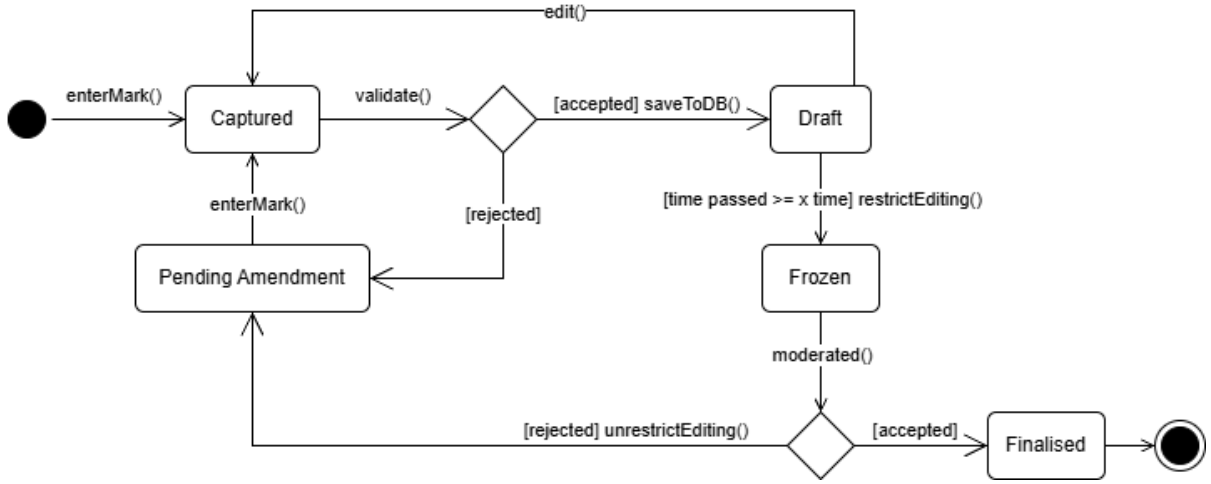


Figure 15 illustrates how a minor register at a facility and enrols in a program, activating their account and keeping it active through academic progression. They can complete or withdraw from the program (causing their account to be archived), or take leave, temporarily freezing their account. Should they return for any reason, their account is restored and reactivated.

Figure 15: SMD depicting student accounts

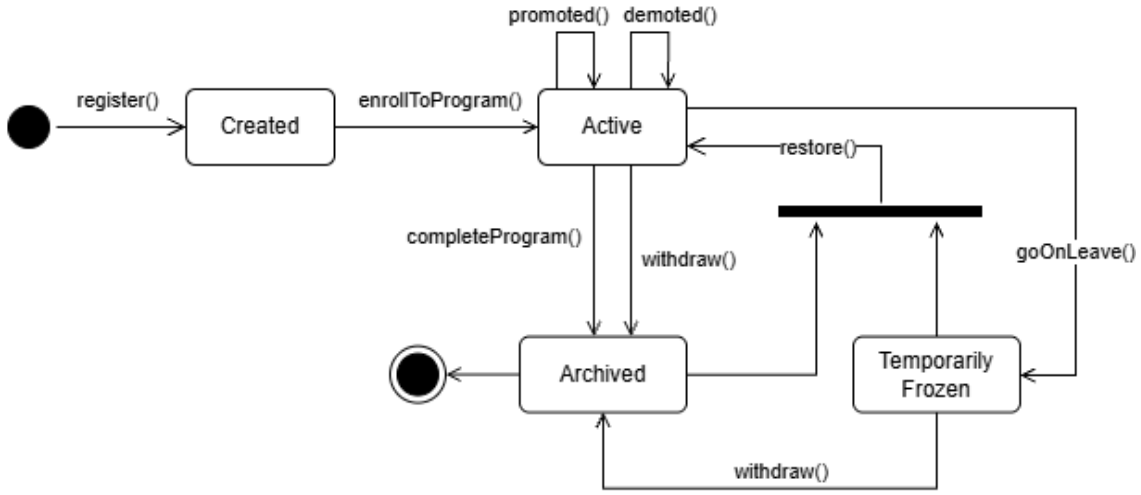
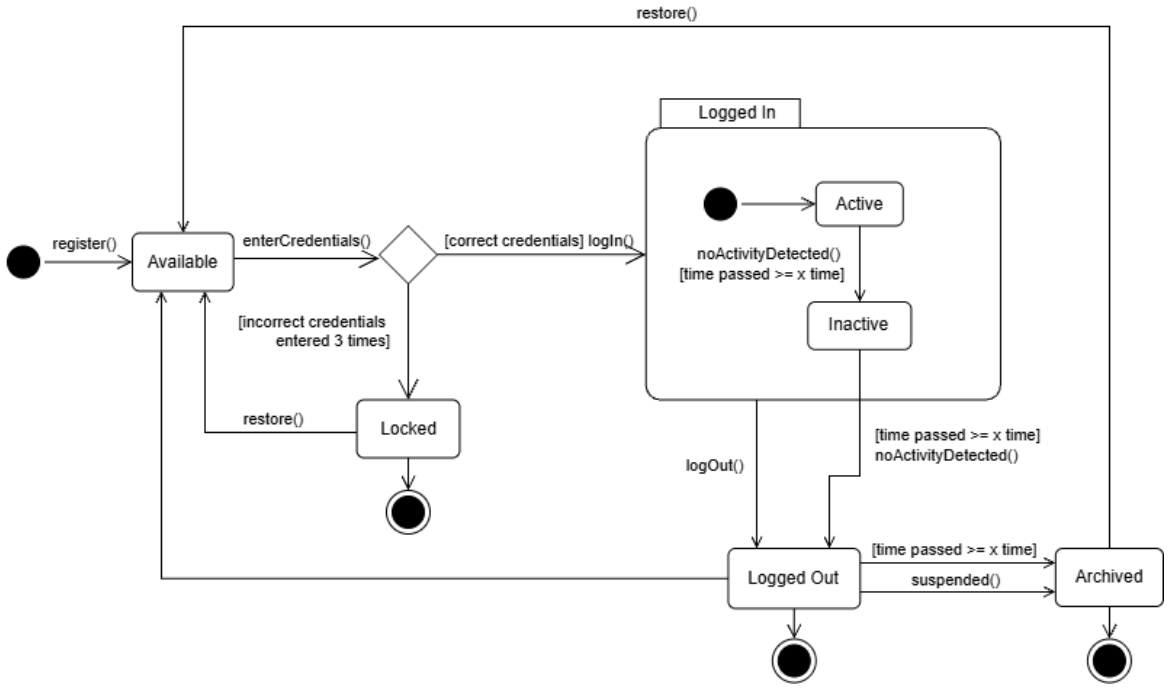


Figure 16 illustrates how non-student users register to create their accounts. They have 3 chances to enter their correct credentials to log in, or the account will be temporarily locked. Inactivity for a prolonged period or manually logging out logs the user out of the account. Suspension or prolonged inactivity archives the account.

Figure 16: SMD depicting user account



Website Navigation

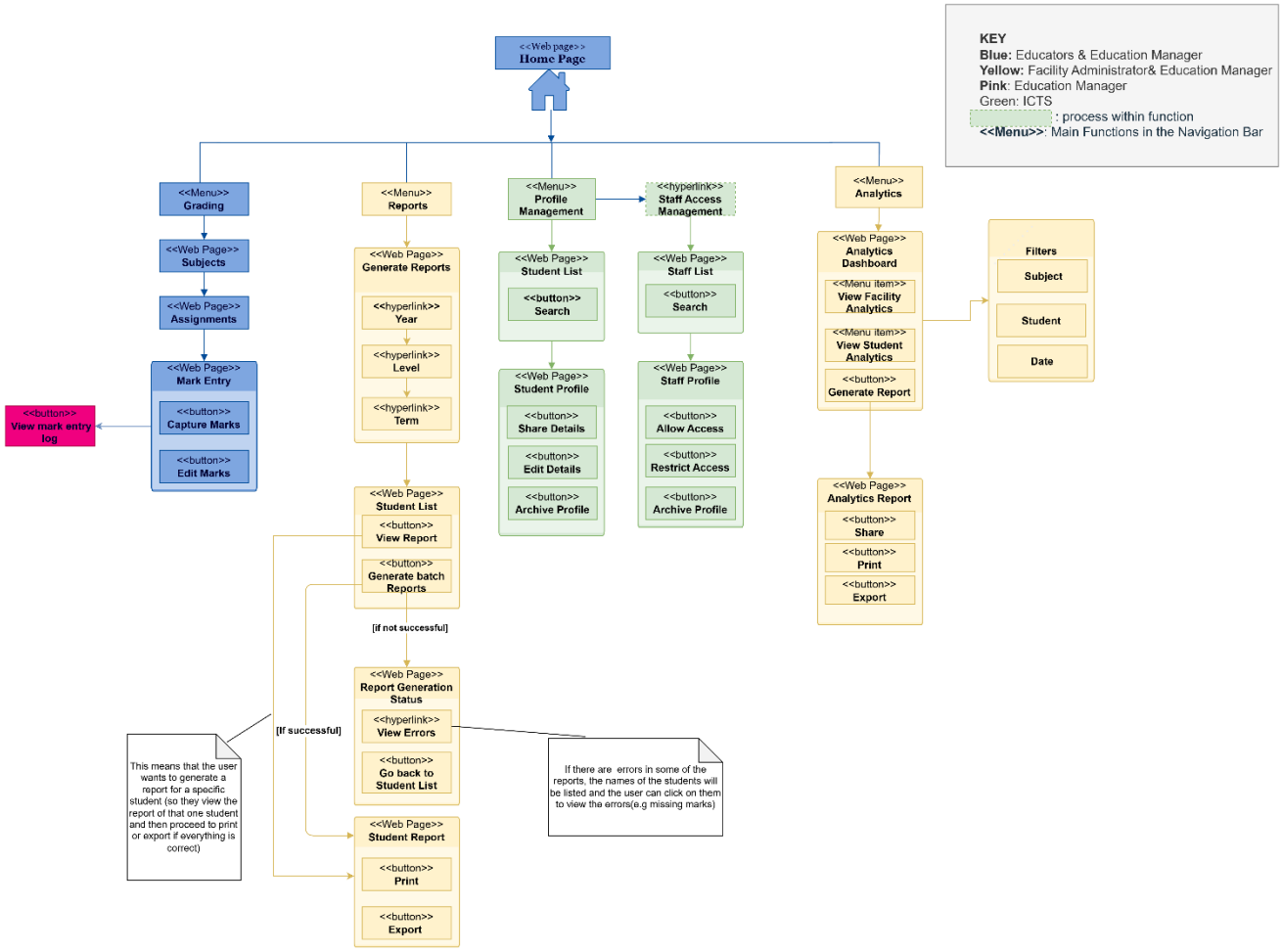


Figure 17: Web Navigation Diagram

The navigation diagram above illustrates how a teacher or administrator navigates the Educational Reporting System after logging in. The system provides four main functions: Grading, Reports, Student Profiles, and Analytics. Through the Grading menu, users select a subject and assignment before accessing a page where student marks can be captured or edited. The Reports menu allows users to select a student, generate a report, and either print or export it. The Student Profiles section enables users to view detailed information about individual students. The Analytics section provides access to facility-wide and student-specific performance data, which can be filtered by criteria such as subject and date. The diagram visually represents the pages, navigation elements, and actions available within the system, showing how users move between different features to complete their tasks efficiently.

Updated Task Board

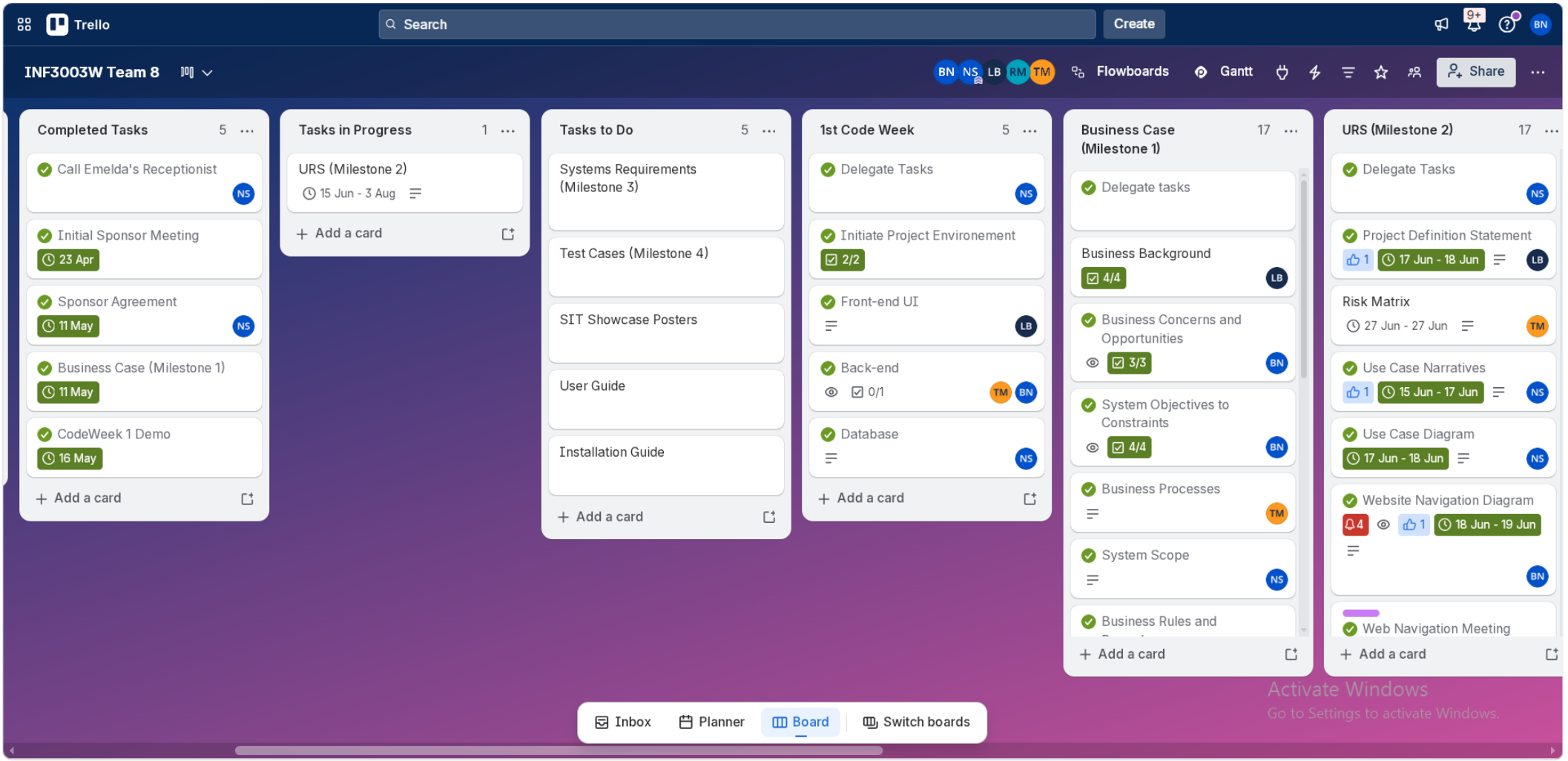


Figure 18: Trello Task board

For Milestone 2, the Trello task board was updated to support the progression of the project beyond the Business Case phase. The board continues to serve as the team's primary project management tool, providing a centralized platform for planning, assigning, monitoring, and reviewing project activities. Trello was selected because it is free, accessible to all team members, and supports real-time collaboration, allowing the team to effectively coordinate work and monitor progress across multiple project phases.

As the project has advanced, the task board has expanded to include milestone-specific lists, enabling the team to track deliverables associated with each project stage. This structure provides greater visibility of project progress, helps ensure deadlines are met, and allows team members to easily identify dependencies between tasks.

Completed Tasks

This list contains all tasks that have been completed and approved. It serves as a record of work accomplished throughout the project and allows the team to monitor overall progress. Examples include stakeholder meetings, project initiation activities, sponsorship approval, completion of the Business Case (Milestone 1), and the first code week demonstration.

Tasks in Progress

This list contains tasks that are currently being worked on by team members. At the time of this milestone, work is focused on the User Requirements Specification (URS), indicating that the project has moved into the requirements analysis phase.

Tasks To Do

This list contains upcoming tasks that have been identified but not yet started. These tasks include future deliverables such as System Requirements, Test Cases, User Guides, Installation Guides, and showcase preparation activities. The list acts as the project's backlog and helps the team plan future work.

1st Code Week

This list tracks development activities completed during the first coding iteration of the project. Tasks include setting up the project environment, assigning development responsibilities, front-end development, back-end development, and database implementation. Keeping these activities grouped together allows the team to monitor technical progress separately from documentation milestones.

Business Case (Milestone 1)

This list contains all activities completed during the Business Case phase of the project. Tasks include the Business Background, Business Concerns and

Opportunities, System Objectives and Constraints, Business Processes, System Scope, Business Rules and Procedures, Feasibility Assessment, and other business case deliverables. Maintaining this list provides a record of all completed milestone activities for future reference.

URS (Milestone 2)

This list contains tasks related to the User Requirements Specification phase of the project. Activities include the Project Definition Statement, Risk Matrix, Use Case Narratives, Use Case Diagram, Website Navigation Diagram, stakeholder meetings, and the To-Be Activity Diagram. This milestone focuses on understanding user requirements and documenting system functionality before detailed design and development proceed.

System Requirements (Milestone 3)

This list has been created to accommodate future system analysis and design activities. It will contain tasks related to defining functional and non-functional system requirements once the URS phase has been completed.

The updated task board provides a clear overview of completed work, current activities, and future milestones, enabling effective project tracking, improved accountability, and better coordination among team members throughout the project lifecycle.

Portfolio of Evidence

This section briefly describes the division of tasks between each member of Team 8, for the completion of this URS.

Nicole Scott-Emuakpor – Project Manager

Category	Description
Role in Project	As Project Manager, Nicole's main role was to create and maintain the project plan aligns with the academic timeline. By scheduling meetings with our sponsor, mentor and tutor and acting as the primary point of contact between the team and the WCG sponsor. Additionally, sent and co-ordinated the sponsor agreement form
Work done in the document	• Package Diagram • Use Case Diagrams • Use Case Narratives • Document Formatting Final Check • Document Content Final Check • Signed AI declaration form
Managing Meetings	• Attended each meeting with sponsor, mentor, and tutor • Set up a meeting via Teams • Set up recurring meetings between tutor and mentor • Aligned meetings with team members' schedules
Presentation	• Explained the business problem and pain points identified during the analysis phase and discussed how the prototype addresses these challenges during code week.
Coding/Prototyping	• Designed and implemented the database structure required to store captured learner information and established the connection between the application and the database during code week. • Developing the Subject Management of the section.
Research	• Researched the recommended tech stack given to us by WCG • Compared the advantages and disadvantages of working with either the Microsoft stack (.Net) or Open Source • Recommended the relevant tools to work on the project

Tsholo Mnguni – Software Developer

Category	Description
Role in Project	Tsholo's role as Software Developer is the actually building and co-ordinating of the EPRS. She assisted in the research for the tech stack of the system and finally co-ordinated and assisted to set up the Trello Board. Tsholo took meeting minutes for each meeting we had with our sponsor and mentor.
Work done in the document	• Risk Assessment • Activity Diagram • State Machine Diagrams • Sequence Diagrams • Document Formatting Final Check • Document Content Final Check • Signed AI declaration form
Managing Meetings	• Attended each meeting with sponsor, mentor, and tutor • Assisted in writing up meeting minutes for all our relevant meetings • Called for additional meetings between team members to discuss delegating work and unclear communication. • Managed the Trello board and set up task completion timelines for deliverables.
Presentation	• Presented the technology stack used during development, including the tools, frameworks, and technologies selected for the prototype.
Coding/Prototyping	• Assisted in implementing the data capture functionality by developing the application logic required to process and manage user input. • Set up the project repository and is responsible for the database implementation and related database functionality.
Research	• Researched the recommended tech stack given to us by WCG • Compared the advantages and disadvantages of working with either Microsoft stack (.Net) or OpenSource • Recommended the relevant tools to work on the project

Bianca Ngobeni – Business Analyst

Category	Description
Role in Project	As the Business Analyst, Bianca is responsible for understanding what WCG needs the system to do and translating that into clear requirements for the development. Additionally, assisted with the Trello Board and its upkeep
Work done in the document	• Website Navigation Diagram • Portfolio of Evidence • Project Task Board • Risk Assessment • Document Content Final Check • Signed AI declaration form
Managing Meetings	• Attended each meeting with sponsor, mentor, and tutor • Assisted in writing up meeting minutes for all our relevant meetings • Attended the additionally meeting with WCG Cel session
Presentation	• Conducted the live walkthrough of the prototype, demonstrating the Data Capture functionality and showing how users interact with the system.
Coding/Prototyping	• Worked alongside the Software Developer on the system functionality, helping to develop and test the data capture processes and ensuring information was correctly handled by the application during code week. • Developing the Student Management section.
Research	• Researched the recommended tech stack given to us by WCG • Compared the advantages and disadvantages of working with either Microsoft stack (.Net) or OpenSource • Recommended the relevant tools to work on the project

Lethiwe Buthelezi – Soft Skills

Category	Description
Role in Project	As Soft Skills, Lethiwe works with the interpersonal qualities amongst our team. She organised our team building session and assists in mitigating any conflict within the team.
Work done in the document	• Project Definition • Class Diagram • Conclusion • Document Content Final Check • Signed AI declaration form
Managing Meetings	• Attended each meeting with sponsor, mentor, and tutor • Assisted in writing up meeting minutes for all our relevant meetings
Presentation	• Introduced the project, provided an overview of the Educational Reporting System, and explained the purpose of the prototype during code week.
Coding/Prototyping	• Designed and developed the user interface using Blazor pages. Created the data capture screens and ensured that the forms were user-friendly and easy to navigate during code week. • Developing Staff Management section.
Research	• Researched the recommended tech stack given to us by WCG • Compared the advantages and disadvantages of working with either Microsoft stack (.Net) or OpenSource • Recommended the relevant tools to work on the project

Conclusion

The URS provided deep understanding of the proposed Educational Reporting System through the analysis of current pain points and system requirements. The project team identified key functionalities required to improve the management of learner academic information within juvenile facilities. The proposed system aims to simplify essential processes such as capturing and editing learner marks, generating report cards, managing student and staff profiles, and providing analytics to keep track of student progress and make informed decisions. The documented diagrams demonstrate how users will interact with the system and how processes will be handled from initiation to completion. The URS also established the foundation for future system development by clearly defining user expectations, system functionality, and the responsibilities of different roles. The requirements gathered during this phase will guide the upcoming design and implementation stages, ensuring that the final system aligns with the operational needs of the Western Cape Government and its educational reporting processes.